

NODIRBEK MAKHTUMOV

AI/ML Engineer · Computer Vision & Edge AI · Video Intelligence

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SUMMARY

AI/ML engineer with 3+ years shipping production computer vision on edge hardware — object detection, segmentation, and real-time video intelligence. Trains and adapts models (YOLO with a custom ResNet50 backbone, SAM2-based auto-labeling), compresses them for NVIDIA Jetson with TensorRT / ONNX quantization and pruning, and delivers the systems around them — FastAPI services, PostgreSQL logging, Dockerized edge deployment. E-7 visa holder, based in Daejeon. B.S. in Computer Science.

CORE SKILLS

Computer Vision & Video: Object detection, semantic / instance segmentation, pose estimation, OCR (EasyOCR); object tracking & multi-object tracking, SAM2 video auto-labeling; diffusion models (fine-tuning)

Model Training & Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, Ultralytics (YOLO), Hugging Face Transformers, OpenCV; experiment tracking — TensorBoard, Weights & Biases, MLflow

Edge & Deployment: Quantization, pruning, TensorRT, ONNX; NVIDIA Jetson (Orin Nano, Orin NX), Raspberry Pi 4 / 5; Docker, FastAPI / REST APIs, Linux (Ubuntu)

Languages & Tools: Python, C++, SQL, PostgreSQL, Git / GitHub, GitLab, VS Code, PyCharm

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Sep 2023 – Present

Netvision Telecom · Daejeon, South Korea · joined as Research Intern (paid), Apr 2023

- Built an edge-aware segmentation pipeline end to end: lightweight segmentation model with RANSAC-based boundary extraction and a custom point-stabilization algorithm for curved edges, served through a FastAPI endpoint with inference metadata logged to PostgreSQL — stable 25+ FPS across 4 concurrent camera streams.
- Developed the vision core of a smart trash-truck management system: YOLO models for trash classification and volume estimation; replaced the YOLOv8 backbone with ResNet50, cutting GPU memory usage ~30% on NVIDIA Jetson Orin NX, and deployed the optimized model for on-truck edge inference.
- Built a SAM2-powered automated video-labeling tool: from a single bounding-box or segmentation prompt, it propagates consistent labels across the entire video and exports annotated images + labels — turning raw footage into efficient training datasets for CV models.
- As research intern (Apr – Sep 2023): built and deployed YOLO vision models on Jetson Orin Nano, applied latency optimization and model quantization, and collected, normalized, and labeled training datasets.

Freelance AI/ML Engineer (Contract)

Smart Truck Parking & Custom Truck-ID Recognition System

- Built and delivered an end-to-end real-time vision system for automated truck-parking monitoring: YOLO detection of trucks and custom truck-ID ROIs, a custom deskewing pipeline with EasyOCR integration, and live occupancy data streamed to the backend — lifting truck-ID recognition accuracy from 76% to 92% in real-world conditions.

Mentor — Woosong University Capstone

- Mentored a student team through a computer-vision capstone (image processing, YOLO model training) with weekly feedback on progress; the team finished 1st among 200+ teams.

EDUCATION

B.S. in Computer Science

2019 – 2023

Woosong University, Daejeon, South Korea

LANGUAGES & VISA

English — Fluent · Korean — Conversational · Russian — Conversational

Visa: E-7 Special Ability (특정활동) — authorized to work in South Korea